

MELBOURNE ESSENDON, AustraliaWMO: **958660**Lat: **37.7275S**Lon: **144.9067E**Elev: **79**StdP: **100.38**Time Zone: **10.00 (AUS)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	4.2	0.0	3.8	14.1	1.1	4.1	11.4	13.1	12.2	12.1	11.9	2.4	0	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	35.6	19.1	32.8	18.5	30.3	18.1	21.4	29.1	20.4	28.2	19.5	26.9	6.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.3	23.1	18.1	13.2	22.2	17.2	12.4	21.6	62.6	29.8	59.1	28.5	56.1	27.0	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.5	9.5	0.3	41.3	1.0	2.3	-0.4	43.0	-0.9	44.3	-1.5	45.6	-2.2	47.3		
			0.0	23.4	0.9	1.1	-0.7	24.1	-1.2	24.7	-1.7	25.3	-2.4	26.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	20.6	20.3	18.9	15.7	12.8	10.4	9.9	10.5	12.3	14.5	17.0	18.8
	DBStd	4.93	4.15	3.51	3.61	3.04	2.41	1.95	1.83	2.18	2.64	3.45	3.94	3.91
	HDD10.0	69	0	0	0	0	3	18	24	19	4	1	0	0
	HDD18.3	1501	17	17	36	90	173	239	260	243	183	128	74	41
	CDD10.0	1937	330	288	276	173	88	29	22	34	73	142	209	272
	CDD18.3	325	89	71	54	12	0	0	0	0	2	11	33	54
	CDH23.3	3552	991	682	519	106	1	0	0	0	10	159	409	676
	CDH26.7	1582	514	294	213	20	0	0	0	0	1	47	161	332
Wind	WSAvg	4.9	5.2	4.9	4.7	4.2	4.6	4.3	5.1	5.1	5.2	5.1	5.0	5.0
Precipitation	PrecAvg	537	43	44	34	46	41	39	36	46	48	55	58	47
	PrecMax	821	102	201	86	142	156	105	94	97	127	144	158	139
	PrecMin	310	4	1	4	5	8	12	7	16	11	16	18	2
	PrecStd	121	24	49	19	33	29	21	20	21	26	32	36	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	36.5	35.9	29.1	22.2	17.6	16.7	20.0	24.5	31.1	34.0	38.4
		MCWB	20.5	19.5	19.3	16.2	13.9	12.4	11.9	13.4	14.7	15.8	17.7	18.8
	2%	DB	36.1	33.2	31.7	26.3	20.0	15.9	15.2	17.7	21.5	27.4	31.0	34.3
		MCWB	19.1	19.1	18.1	15.7	13.9	11.9	11.2	11.9	13.9	15.6	17.5	18.3
	5%	DB	32.3	30.4	28.8	23.9	18.5	14.9	14.2	16.1	19.5	24.6	28.0	30.2
		MCWB	19.1	18.8	17.7	15.3	13.4	11.4	10.5	11.1	12.9	14.8	17.1	17.5
	10%	DB	28.6	27.4	25.8	21.9	17.2	13.8	13.3	14.6	17.7	21.8	25.1	26.8
		MCWB	18.4	18.7	17.2	14.8	12.7	11.0	9.9	10.3	11.9	14.0	16.8	17.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	22.2	21.2	18.3	16.2	13.6	13.2	13.8	15.8	18.3	21.0	21.7
		MCDB	29.6	29.9	28.6	25.3	19.7	15.5	15.7	19.1	21.7	25.4	28.8	29.2
	2%	WB	21.5	21.1	19.9	17.2	14.7	12.7	11.9	12.4	14.5	16.7	19.1	19.9
		MCDB	29.9	28.0	26.9	22.3	18.4	14.7	14.2	16.6	20.2	24.3	25.7	28.2
	5%	WB	20.5	20.1	19.1	16.4	13.9	12.0	11.0	11.5	13.5	15.6	18.2	18.9
		MCDB	28.9	27.1	25.0	21.2	17.5	14.4	13.4	15.1	18.6	22.6	25.0	26.7
	10%	WB	19.4	19.2	18.2	15.5	13.1	11.3	10.3	10.7	12.4	14.4	17.1	18.0
		MCDB	27.5	25.8	24.0	20.4	16.3	13.5	12.7	14.1	16.8	20.7	24.3	25.2
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.4	10.1	9.3	7.5	6.6	6.3	7.3	8.9	10.5	10.5	11.4
		MCDBR	18.9	17.4	16.0	13.7	9.3	8.1	7.2	9.2	12.0	16.3	16.9	19.2
		MCWBR	5.4	5.6	5.4	5.6	5.1	5.0	4.4	5.0	6.2	6.8	5.8	6.1
	5% WB	MCDBR	14.3	12.7	11.6	10.6	8.0	6.8	6.1	8.3	10.7	13.8	13.4	14.2
MCWBR		5.3	5.3	4.9	5.1	4.8	4.5	4.1	4.9	5.9	6.5	5.6	5.6	
Clear-Sky Solar Irradiance	taub	0.349	0.354	0.329	0.330	0.311	0.301	0.301	0.313	0.332	0.335	0.347	0.340	
	taud	2.509	2.503	2.572	2.537	2.574	2.592	2.583	2.534	2.478	2.491	2.484	2.523	
	Ebn at Noon	984	955	941	875	834	816	839	881	921	964	979	997	
	Edh at Noon	112	108	93	84	70	64	68	83	100	108	114	111	
All-Sky Solar Radiation	RadAvg	7.17	6.25	5.00	3.46	2.29	1.77	1.98	2.80	4.00	5.34	6.29	7.05	
	RadStd	0.45	0.33	0.29	0.21	0.14	0.14	0.12	0.19	0.29	0.43	0.38	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	+0.50	N/A	N/A	+0.87	N/A	N/A	-17	-127	+170	+58

Nomenclature: See separate page